

Network access controls — security groups, endpoint policies, and the layers between

Companion to [architecture.md](#), answering one question in depth: **who is allowed to talk to what, and which mechanism enforces it**. Three visuals carry the story; the tables give the exact rules. Diagrams are generated by [diagrams/generate.py](#) — regenerate rather than hand-editing the SVGs.

Accepted risks, up front (SSP-scoped decisions, not oversights):

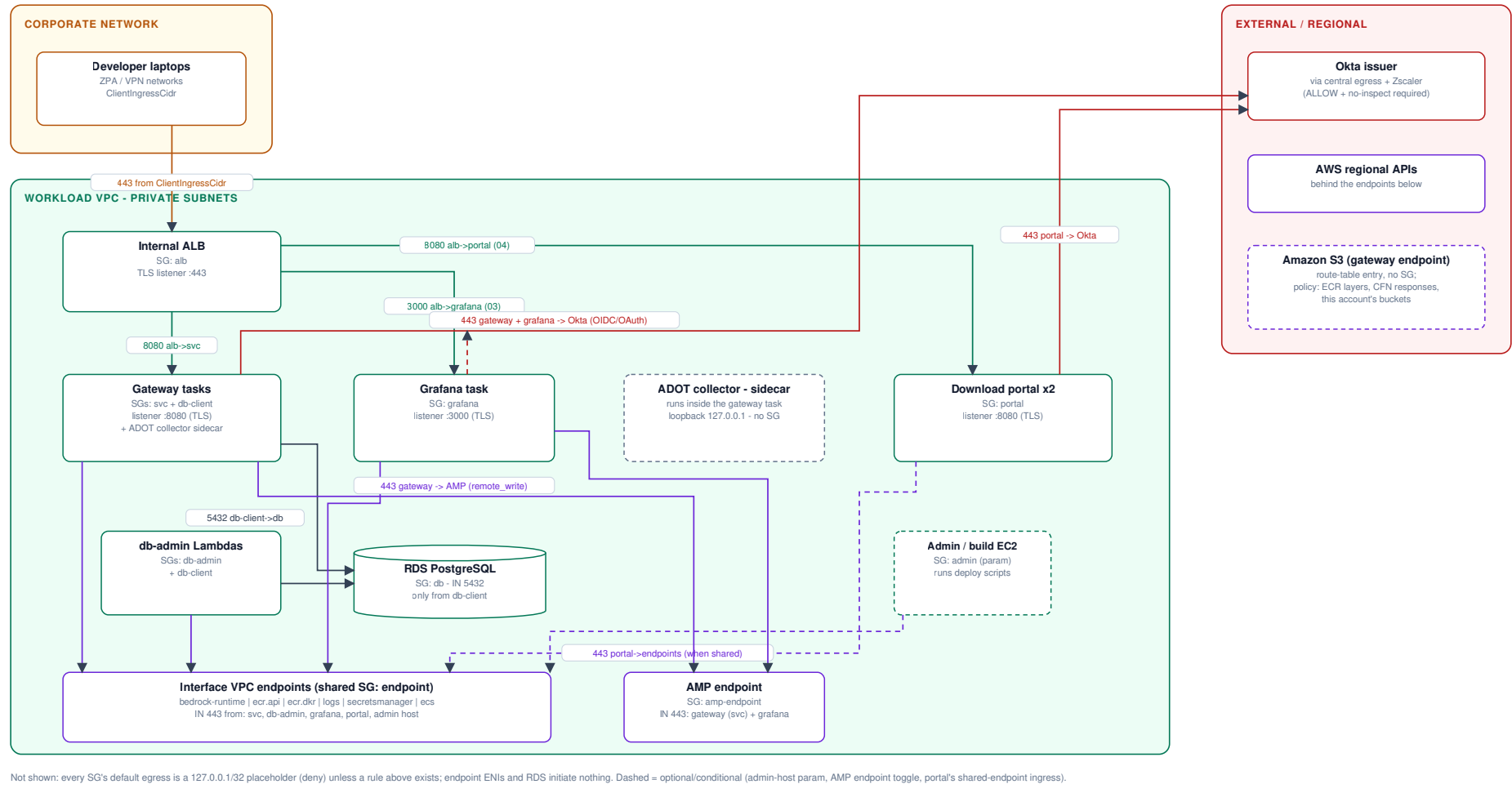
- The `ecs` interface endpoint carries **no endpoint policy** — GovCloud does not support one there; IAM-side scoping covers it.
- The ALB access-logs bucket is SSE-S3, not CMK — ELB log delivery does not support KMS.

1. The map — who may talk to whom

Every arrow is an explicit SG rule *pair*: an egress rule on the source group **and** an ingress rule on the destination group. A connection with only one half configured does not work; a connection with neither is dropped silently. Every group's default allow-all egress is removed, but not all the same way: `alb`, `db-client`, `db`, `endpoint`, and `amp-endpoint` suppress it with a `127.0.0.1/32` placeholder rule (their real egress, where any exists, is added as a separate standalone rule to break a reference cycle, or there is genuinely none); `svc`, `db-admin`, `grafana`, and `portal` instead declare their real 443 egress inline, with no placeholder rule at all. Either way, nothing here is reachable "by default."

7. Security groups - who may talk to whom

Every arrow is an explicit SG rule pair (egress on the source + ingress on the destination). Everything else is denied.



Reading tips:

- **Workloads carry more than one SG.** The gateway tasks run with `svc + db-client`; the db-admin Lambdas with `db-admin + db-client`. Attaching 01's `db-client` group is *the* mechanism for "may reach the database" — nothing else is ever granted 5432.

- **The interface endpoints are the chokepoint.** Their private DNS captures AWS API calls from **every** VPC client, so any in-VPC caller of ecr/logs/secretsmanager/ecs — including the admin/build host and tasks created by a different stack — must appear in the endpoint SG's ingress or its calls black-hole. This is the most common cause of a downstream stack rolling back at image pull or secret read.
- **Okta is the only external dependency** the workloads dial out to; it rides the landing zone's central egress and needs the Zscaler ALLOW + no-inspect prerequisite (see `networking-request-email.md`).

2. The inventory — every rule, by group

8. Security-group rule inventory

Every rule in the deployment, by group. IN = ingress, OUT = egress. Groups not listed here do not exist.

alb (internal ALB)

02

IN 443 from ClientIngressCidr (developer networks)
OUT 8080 to svc (forward + health checks)
OUT 3000 to grafana (rule added by 03)
OUT 8080 to portal (rule added by 04)

svc (gateway tasks)

02

IN 8080 from alb
OUT 443 0.0.0.0/0 (Okta, AWS APIs via endpoints/egress)
OUT proxy port 0.0.0.0/0 (only when HttpsProxyUrl set)
OUT 443 to amp-endpoint (remote_write; rule added by 03)
also carries db-client (5432) + telemetry sidecar (loopback)

db-admin (bootstrap + rotation Lambdas)

02

IN none
OUT 443 0.0.0.0/0 (Secrets Manager, ECS via endpoints)
also carries db-client (below) for 5432

endpoint (shared, all 02 interface endpoints)

02

IN 443 from svc, db-admin
IN 443 from grafana (rule added by 03)
IN 443 from portal (rule added by 04, when shared)
IN 443 from AdminClientSecurityGroupId (optional param)
OUT none (endpoint ENIs never initiate)

portal (download portal task)

04

IN 8080 from alb (path rule /portal, /portal/*)
OUT 443 0.0.0.0/0 (Okta, S3, CloudWatch, ECR, Secrets Mgr)
OUT proxy port 0.0.0.0/0 (only when HttpsProxyUrl set)

db-client (attach to reach the DB)

01

IN none
OUT 5432 to db
attached to: gateway tasks, both db-admin Lambdas

db (RDS instance)

01

IN 5432 from db-client
OUT none

grafana (dashboard task)

03

IN 3000 from alb (path rule /grafana)
OUT 443 0.0.0.0/0 (AMP queries, Okta OAuth, ECR)

amp-endpoint (aps-workspaces endpoint)

03

IN 443 from svc (gateway task - remote_write)
IN 443 from grafana (queries)
IN 443 from AdminClientSecurityGroupId (optional param)
OUT none

Cross-stack rule writers

03 adds rules to SGs it imports from 02:
- AlbToGrafanaEgress: alb OUT 3000 -> grafana
- GatewayToAmpEndpointEgress: svc OUT 443 -> amp-endpoint
- GrafanaToEndpointsIngress: endpoint IN 443
- AdminToAmpEndpointIngress: amp-endpoint IN 443
04 adds rules to SGs it imports from 02:
- AlbToPortalEgress: alb OUT 8080 -> portal
- PortalToEndpointsIngress: endpoint IN 443 (when shared)
02 param AdminClientSecurityGroupId:
- AdminToEndpointsIngress: endpoint IN 443

Deploy order matters: imported SGs must exist first
(02 before 03/04); CREATE_SUPPORTING_ENDPOINTS
must match across the deploys.

Reading the map

A connection works only when BOTH ends agree:
an egress rule on the source SG and an ingress
rule on the destination SG. Every SG here is
default-deny; the 127.0.0.1/32 egress entries in
the templates are deliberate no-op placeholders
that suppress the default allow-all egress.

The gateway->collector OTLP hop is not on the network - the ADOT collector runs as a co-resident sidecar in the gateway task, reached over loopback (127.0.0.1:4318). No SG rule exists or is needed for it.

SG	Stack	Attached to	Ingress	Egress
alb	02	internal ALB	443 from ClientIngressCidr ; 443 from portal (04, unconditional — every portal task is an ALB <i>client</i> : the fingerprint and self-usage pages need it, not only the admin pages)	8080→ svc ; 3000→ grafana (03); 8080→ portal (04)
svc	02	gateway tasks (incl. the co-resident ADOT collector sidecar)	8080 from alb	443 anywhere; proxy port (optional); 443→ amp-endpoint (03)
db-client	01	gateway tasks, db-admin Lambdas	—	5432→ db
db	01	RDS instance	5432 from db-client	—
db-admin	02	bootstrap + rotation Lambdas	—	443 anywhere
endpoint	02	all 02 interface endpoints	443 from svc , db-admin , grafana (03), portal (04, when 02 created the shared endpoints), admin host (param)	—
grafana	03	Grafana task	3000 from alb	443 anywhere
amp-endpoint	03	aps-workspaces endpoint	443 from svc (gateway sidecar), grafana , admin host (param, when granted)	—
portal	04	download-portal tasks	8080 from alb	443 anywhere (Okta, S3, CloudWatch); proxy port (optional)

There is **no collector security group**: the ADOT collector is a co-resident sidecar inside the gateway task, reached over loopback (127.0.0.1:4318), so its OTLP receiver is never on the network and needs no SG rule — which is why there is no gateway→collector network hop to encrypt (architecture.md §10 note 1). The sidecar's remote-write to AMP rides the gateway svc SG.

Cross-stack rule writers (03, 04 and the admin-host parameter modify imported 02 SGs — or, for AdminToAmpEndpointIngress , 03's own amp-endpoint SG — as separate SecurityGroup{In,E}gress resources): AlbToGrafanaEgress , GrafanaToEndpointsIngress , GatewayToAmpEndpointEgress (the sidecar's remote-write path), AdminToEndpointsIngress , AdminToAmpEndpointIngress (admin host → amp-endpoint , same param, mirrors AdminToEndpointsIngress); from 04: AlbToPortalEgress , PortalToAlbIngress (unconditional — the fingerprint page and /portal/me self-usage make every portal task an ALB client, not only admin-enabled ones), PortalToEndpointsIngress (only when 02 created the shared endpoints). Deploy order (02 before 03/04) and a matching CREATE_SUPPORTING_ENDPOINTS across the deploys are what make these land correctly.

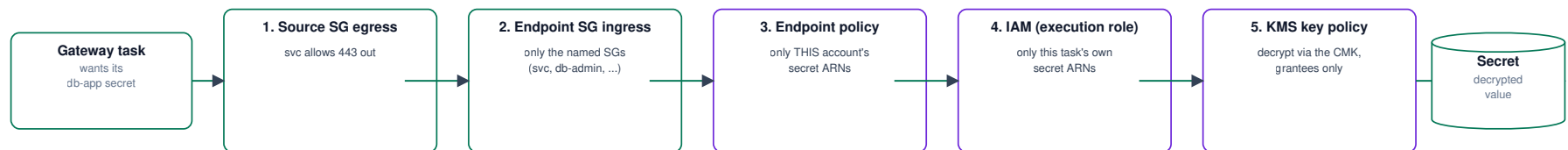
3. The layers — what each control stops that the others can't

SGs are one of **five** gates on the path of a single AWS API call. The diagram traces a gateway task reading its DB secret; the failure table under it is the reason none of the layers is redundant — the second row (attacker-owned credentials exfiltrating *through* a private endpoint) is the one SGs and your own IAM cannot

stop alone, and is why every endpoint that **supports** a resource policy carries one. (The one deliberate exception is `ecs` — GovCloud does not support a policy on that endpoint type; IAM-side scoping is the only gate there, per the accepted-risk note above.)

9. Layered access control - what each layer stops

One request traced through every gate: a gateway task reading its DB secret through the Secrets Manager endpoint.



Why the layers are not redundant - each stops a different failure:

- **Compromised box in another subnet, ANY credentials**
stopped at layer 2 - its SG is not in the endpoint SG ingress (network path denied before authn)
- **Compromised workload using ATTACKER-OWNED AWS credentials (exfiltration through the private endpoint)**
stopped at layer 3 - the endpoint policy allows only this account's resources; your IAM cannot judge foreign principals
- **Legitimate workload asking for a secret it does not own**
stopped at layer 4 - execution roles enumerate exact ARNs (gateway task cannot read the master secret: no task injects it)
- **Principal with a stolen ciphertext or over-broad S3/API access**
stopped at layer 5 - the CMK key policy grants decrypt only to the roles that need it

Non-SG enforcement points elsewhere in the deployment

bedrock-runtime endpoint policy: pinned to the THREE configured model IDs + their inference profiles (not anthropic.*) | ecs endpoint: NO policy (GovCloud unsupported) - IAM-side scoping covers it
S3 gateway endpoint policy: ECR layer bucket + CloudFormation custom-resource response buckets + this account's buckets only
ALB access-log bucket policy: exactly two writers (ELB log-delivery service principal + legacy regional ELB account) | AMP endpoint policy: this workspace's ARN only
Okta egress: Zscaler policy must ALLOW + not inspect the issuer FQDN for server-originated traffic (org prerequisite - see the networking request)

Endpoint/resource policies in force:

Where	Policy scope
<code>bedrock-runtime</code> endpoint	the three configured model IDs + their inference profiles — not <code>anthropic.*</code>
<code>ecr.api</code> / <code>ecr.dkr</code> endpoints	auth token (no-resource) + this account's repositories
<code>logs</code> endpoint	this account's log groups
<code>secretsmanager</code> endpoint	this deployment's secret ARNs + <code>GetRandomPassword</code>
<code>ecs</code> endpoint	none (unsupported in GovCloud) — IAM covers
S3 gateway endpoint	ECR layer bucket, CloudFormation custom-resource response buckets, this account's buckets
<code>aps-workspaces</code> endpoint (03)	this AMP workspace's ARN only
ALB access-logs bucket	exactly two writers: ELB log-delivery service principal + legacy regional ELB account

IAM and KMS (layers 4–5) are inventoried in `architecture.md` §6 — the short version: execution roles enumerate exact secret ARNs, the task role holds `bedrock:InvokeModel*` on exactly the three models, and the single CMK's key policy names its grantees.

Change discipline: a new workload that needs the database gets `db-client` attached (never a new 5432 rule); a new in-VPC caller of AWS APIs gets an ingress on the `endpoint` SG (or it will black-hole); anything touching the `endpoint` SG or the observability SGs should re-check cross-stack reachability by hand — no static gate catches it (`../operations/troubleshooting.md`).